

# EEG Signal Preprocessing

What raw scalp EEG actually is, and why we clean it the way we do — CHB-MIT Scalp EEG Database

## 1. What you're actually looking at

A raw EEG recording is simply a set of voltage measurements over time, one stream per electrode. In CHB-MIT, each recording is stored as a matrix of shape (n\_channels, n\_samples): one row per channel, one column per moment in time. Everything in preprocessing exists to turn this noisy, messy matrix into something a model — or a human — can actually interpret.

## 2. Sampling rate

The sampling rate (256 Hz for CHB-MIT) tells you how many voltage measurements are taken per second, per channel. This single number governs almost every other calculation in the pipeline: converting a time window (in seconds) into a number of samples always means multiplying by the sampling rate. A 10-second window at 256 Hz is 2,560 samples long — not because of any special property of EEG, just arithmetic. If you ever process data recorded at a different rate (say, 500 Hz clinical equipment), every downstream calculation involving 'how many samples is X seconds' must be rebuilt with the new rate, or every timing calculation becomes silently wrong.

## 3. Montage: how electrodes are actually compared

EEG can be recorded two main ways. In a **referential montage**, every electrode's voltage is measured against one shared reference point. In a **bipolar montage** — what CHB-MIT uses — each channel is instead the *difference* between two adjacent electrodes (e.g. FP1-F7 means 'voltage at FP1 minus voltage at F7').

The practical reason to prefer bipolar recordings: if the reference electrode itself picks up noise (muscle movement, a loose contact), that noise appears in *every single channel* in a referential setup, because they all subtract the same contaminated reference. In a bipolar montage, noise common to two neighboring electrodes cancels out in the subtraction, while the genuine difference in brain activity between them survives. This is why CHB-MIT's channel names look like differences (FP1-F7, F7-T7, T7-P7...) rather than single electrode labels.

## 4. Filtering: what to keep, what to throw away

Raw EEG contains far more than brain activity — sweat, electrode drift, and muscle contraction all leave a mark on the signal. Filtering is how we discard the parts of the signal we know aren't useful, based on their frequency.

- **Band-pass filter (1–40 Hz):** below ~1 Hz is almost always slow drift, not neural activity. Above ~40 Hz, scalp EEG is dominated by muscle artifact, not brain signal. Seizure-related activity lives comfortably inside this range, so the band-pass keeps what matters and discards the rest.
- **Notch filter (60 Hz for CHB-MIT):** removes electrical power-line interference. This number is not universal — it depends on the electrical grid where the recording was made. CHB-MIT was recorded in Boston (60 Hz grid). Clinical data recorded in Egypt would need a 50 Hz notch filter instead. Using the wrong notch frequency doesn't crash anything — it silently leaves line noise sitting in the 'cleaned' signal, which is a genuine engineering trap, not a cosmetic detail.

## 5. Epoching: cutting a long recording into meaningful pieces

A single CHB-MIT recording can span an hour or more, while a seizure itself might last less than a minute. Rather than handing a model an hour of mostly-irrelevant signal, we cut out short windows ("epochs") around the events we actually care about — in this case, seizures. A small padding (e.g. 5 seconds before onset, 5 seconds after the end) is included deliberately, so that a model eventually has a chance to learn what the transition *into* and *out of* a seizure looks like, not only its most obvious middle section.

## 6. Real datasets are not clean — four quirks worth remembering

Working through this dataset first-hand surfaced several genuine irregularities. These aren't bugs in your code — they're properties of real, hospital-collected data that any clinical dataset will eventually show you.

- **Duplicate channel names:** some files genuinely contain two channels both labeled T8-P8. Loading software (MNE) silently renames them (T8-P8-0, T8-P8-1) to keep them distinct, which then breaks any code that expects the exact original name — a fix has to explicitly account for this.
- **Inconsistent channel counts across patients:** most CHB-MIT patients have 23 channels, but some have 24, 28, even 38 — extra channels recorded for that particular patient's clinical needs. A model or feature extractor built assuming a fixed number of channels will break the moment it meets a patient recorded differently.
- **A genuinely different montage (patient chb12):** a subset of one patient's files were recorded with an entirely different electrode setup from the rest of the dataset — not just a different count, a structurally different montage. Published research on this dataset commonly excludes these files outright rather than trying to reconcile them.
- **Irregular filenames:** a recording that had to be split mid-session is marked with a '+' in its filename (e.g. chb02\_16+.edf) — a real, documented convention, not a typo or corrupted download.

## Why this matters beyond this project

Every one of these quirks reflects something true of real clinical data in general, including whatever scalp or intracranial EEG you'll eventually work with for your own PhD dissertation: hospitals don't record data with a research pipeline in mind. Learning to expect, detect, and handle this kind of irregularity — rather than assuming a dataset will behave like documentation says it should — is itself a core, transferable research skill, arguably as valuable as any single filtering formula.